

Bayesian applications in pharmaceutical statistics

1 | INTRODUCTION

I am an untypical pharmaceutical statistician in that my early training in the subject took place in a Bayesian environment, due to spending too much time at an impressionable age (in my Mathematics degree) with Adrian Smith and Bruce Turnbull. But I then moved on to postgraduate work in Biomathematics under Maurice Bartlett, who seemed to regard Bayesian arguments as of academic curiosity only. A decade in agricultural research in the shadow of Fisher and Yates saw not a trace of a Bayesian analysis.

But I had in the back of my mind that Bayesian methodology offered as a minimum a useful model of a decision-making process. So when I moved into the pharmaceutical industry in 1986, I wondered if I would see any more of Revd Bayes' legacy—given the number of decisions that the industry was obviously making, particularly in Phase II with go/no go decisions and dose selection. There was some Bayesian activity around that time with the paper by Amy Racine, Andy Grieve and colleagues from Ciba-Geigy attracting attention both within and outside the industry.¹ But in practice, little use was made of Bayesian techniques—with the prevailing view in Drug Development that the regulatory authorities did not encourage further uptake.

Over the intervening years, several factors have encouraged further adoption of Bayesian methodology. Work has continued within the industry with the torch carried by statisticians including Karen Price in US and Andy Grieve in Europe: Andy reflected on his experiences in this journal.² Software has become more readily available, with few statisticians now completing their studies without some use of WinBUGS or OpenBUGS for example. The development of adaptive trials has involved much Bayesian analysis. Numerous text-books and academic papers have been published over the decades, but little in the way of easy-to-use case studies has been available for busy statisticians to see how they can apply Bayesian methods in their routine (and not-so-routine) work.

2 | REGULATORY AND PROFESSIONAL APPLICATION

In the regulatory area, it is encouraging to note the inclusion of the word “Bayesian” in the latest iteration of the Prescription Drug User Fee reauthorization, which guides FDA operations from 2018–2022. Specifically, FDA's PDUFA VI commitment letter³ states in the section “Enhancing Regulatory Decision Tools To Support Drug Development And Review”:

“FDA will develop the staff capacity to enable processes to facilitate appropriate use of these types of methods. This staff will support the computationally intensive review work necessary to evaluate complex adaptive, Bayesian, and other novel clinical trial designs, with a particular focus on clinical trial designs for which simulations are necessary to evaluate the operating characteristics.”

This development extends the existing recognition of the value of Bayesian approaches within FDA's Center for Devices and Radiological Health.^{4,5} The European Medicines Agency is making tentative steps in this direction also, with acknowledgement of Bayesian approaches in extrapolation of data into paediatric populations.^{6,7}

This journal has played its part in disseminating useful guidance—the paper by O'Hagan et al⁸ was recognised at the 2014 PSI Conference as the most influential paper in the first 10 years of Pharmaceutical Statistics. Walley et al⁹ described the advantages of basing decision-making in early development on a Bayesian paradigm. A special issue on Bayesian Statistics was published in 2014 with coverage of a variety of areas, including network meta-analysis in drug

safety.¹⁰ The issue also contained a paper by the DIA Bayesian Scientific Working Group reporting a survey showing that “having access to fully worked case examples was considered the most helpful factor to start applying Bayesian approaches.”¹¹ In recent years, we have encouraged Andy Grieve to share his knowledge and experience with case studies which are easy to follow and can hopefully cast light on a variety of areas of work which can be improved by a Bayesian approach.^{12,13}

3 | PAPERS IN THIS SPECIAL ISSUE

In my 2+ years as Joint Editor-in-Chief, the journal has seen the proportion of Bayesian papers increasing—even to the point where we were struggling to find enough well-informed reviewers to assess them—and we should no longer have any illusions that Bayesian Applications are a “special topic”—they are really quite mainstream. But the trigger for developing this Special Issue was the offering of two papers from within GlaxoSmithKline,^{14,15} showing how they have found the Bayesian concept of Assurance valuable in their early-phase work and including case studies which show the practical aspects of implementation in their environment. We are grateful for GSK for opening up their processes to this degree and to the authors for describing it in an easy-to-read but rigorous way.

To these two core papers we have added five others which show the variety of ways in which Bayesian methodology can be and is being applied within our industry, including the regulatory process.

Weber et al¹⁶ consider the difficulty of drawing conclusions from the small numbers of patients in paediatric and other special populations in rare diseases. They explore the use of data-based Bayesian meta-analytic-predictive methods to obtain formal proof of efficacy of a new drug by using additional information expressed through a prior distribution.

Ghosh et al¹⁷ address the problem of assessing non-inferiority in a three-arm trial with a binary endpoint. The use of the Bayesian paradigm provides a path to integrate historical trials and current trial information. The authors employ several different methods (both approximate and exact) to evaluate the feasibility of the proposed approach.

Kharroubi et al¹⁸ look at the modelling of health related quality of life data, which is a difficult area as data distributions are heavily skewed with a significant proportion of observations at the extreme of the distribution. They use data from a UK Medical Research Council myeloma trial to examine the relationship between different measures of quality of life and to compare four different two-part approaches to model the data.

Takeda and Morita¹⁹ consider the problem of incorporating historical data from a preceding trial to design and conduct a subsequent dose-finding trial in a different population of patients, for example to bridge oncology results in Caucasian patients across to Asian patients. Their approach incorporates historical data into prior distributions informing a new trial.

Takeda et al²⁰ seek to address the issue that dose-finding methods based purely on toxicity may be inappropriate for non-cytotoxic therapies due to the assumption that both efficacy and toxicity increase monotonically with dose. They extend the model-assisted non-parametric Bayesian optimal interval (BOIN) design of Lu and Yuan²¹ to identify an optimal dose based on both efficacy and toxicity, by minimising the posterior probability of inappropriate dose assignments.

4 | CONCLUSION

I have thoroughly enjoyed putting this issue together – as well as giving insight to new approaches to important problems it has reminded me of a number of half-forgotten topics from my 1970s education! I trust that you will enjoy it too while learning from it and maybe using it to identify the areas in your practice which might benefit from a fresh approach.

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